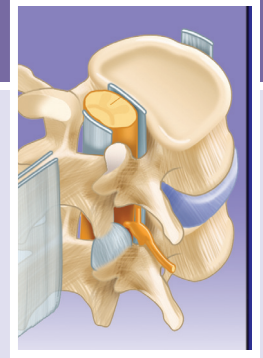


Head and Spine Diagnosis and Decision-Making

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HEAD

Anatomy and Biomechanics

Head injuries have multiple mechanisms, which include the injuries from an impact of the brain against the inside of the skull (Fig. 123.1). A coup injury occurs when the brain strikes the inner skull on the same side of impact of an object and typically occurs when a moving object comes into contact with a patient's stationary head. A countercoup injury occurs when the brain strikes the inner side of a skull on the opposing side of impact of an object, which generally occurs when a mobile patient's head strikes a stationary object. The brain is suspended in cerebrospinal fluid (CSF), a fluid denser than the brain itself. In the cases of countercoup injuries, the denser CSF moves towards the site of impact, likely displacing the brain in the opposite direction.⁴ The mechanisms behind coup and countercoup injuries are only relevant when there is no skull fracture present, because any such displacement to the skull would cause the bone to directly damage brain tissue upon an impact.

Damage to the brain from a head injury may result from three types of applied forces: compressive, tensile, and shearing. Compression force involves squeezing of brain tissue, and tensile force is the opposite of compression and involves stretching of brain tissue. When rotational forces are applied to the brain, shearing stress results from areas where rotational motion of the brain is hindered, such as at rough, irregular contacts between the brain and skull. In severe traumatic brain injuries, brain tissue experiences compression, tension, and shearing. Compressive and tensile forces are handled relatively well by the head because CSF is able to absorb and convert focal shock into a more widely distributed stress throughout the head. However, shearing stresses are less tolerated by the brain, with the most vulnerable parts being the brainstem and corpus callosum due to their highly anisotropically oriented axons.⁷

Pathophysiology

Head injuries are found in a wide variety of sports. Concussion is the most commonly sustained head injury in athletes. However, it is poorly defined, and treatment modalities vary greatly and are dependent on a multitude of factors. Intracranial hemorrhage is the leading cause of death from head injury in athletes; four types exist—epidural, subdural, intracerebral, and subarachnoid. Intracranial hemorrhage is a focal brain injury that results from

a ruptured blood vessel within the skull and leads to dangerous increases in intracranial pressure. Other severe, although less common, injuries include skull fractures, mild and severe traumatic brain injuries, CSF leaks, craniofacial injuries, and cranial nerve injuries.

As mentioned previously, concussions are the most common head injuries occurring in athletes. They have previously been defined as low-velocity injuries resulting in “brain shaking,” and clinical symptoms that may not necessarily be related to the initial pathologic injury ensue.⁸ The most recent (2012) Conference on Concussion in Sport differentiated concussion from a minor traumatic brain injury (mTBI) and considers the concussion to instead be its own subset of TBI.⁸ Concussions can be the result of low-velocity impact to the head, face, neck, or anywhere else in the body that can transmit “impulsive” force to the head. For adults, this is most commonly seen in vehicular collisions and falls, but for children, the most common causes are bicycle and sporting accidents.⁹ Concussions from vehicular collisions are likely to be much more severe and may include loss of consciousness (LOC) as compared with those concussions resulting from sporting accidents. Although concussions commonly occur in athletes, they are considered to be grossly underreported.^{10,11} There are a variety of reasons for which this can occur: a player might believe that reporting a concussion might curtail his player career; a team or organization might not want to report injuries so as not to harm its public relations. Nevertheless, it is important that players, teams, and families fully understand the ramifications of concussions, and it has been demonstrated that an increased knowledge of concussion symptoms is associated with an increase in concussion reporting.^{12,13}

At the time of injury, impairment of neurologic function can occur, although this is not necessary for diagnosis and impairment can appear hours after initial insult. Symptoms of a concussion typically resolve spontaneously. However, the course of recovery is variable, with some concussion patients experiencing symptoms weeks to months after acute injury. Concussions may or may not result in an LOC. In addition, if LOC does occur, it bears no effect on severity of concussion or recovery time. In some cases, concussions can result in the inability to remember new events (antegrade amnesia) or events immediately preceding the concussion (retrograde amnesia). Retrograde amnesia usually resolves spontaneously, although antegrade amnesia from concussions can be permanent.¹⁴ Because the acute clinical symptoms

Abstract

This chapter highlights the recent literature on the diagnosis, decision-making, and treatment options for head- and spine-related injuries. Head and spine injuries are common in a wide variety of sports. Concussion is the most commonly sustained head injury amongst athletes. Common causes of spinal cord injuries include motor vehicle accidents, falls, violence, and sports-related accidents. A detailed and comprehensive history and physical exam is necessary for diagnosing and triaging head and spine injury. Pertinent diagnostic imaging modalities include computed tomography or magnetic resonance imaging for ruling out emergencies, such as intracranial hemorrhage, if necessary. Plain radiography is particularly helpful in spine injuries for identifying fractures and assessing the need for further advanced imaging. Decision-making with head and spine injuries should follow established society guidelines, severity of injury, and results of diagnostic imaging.

Keywords

head injuries
concussion
intracranial hemorrhage
spinal cord injury
catastrophic cervical spine injuries

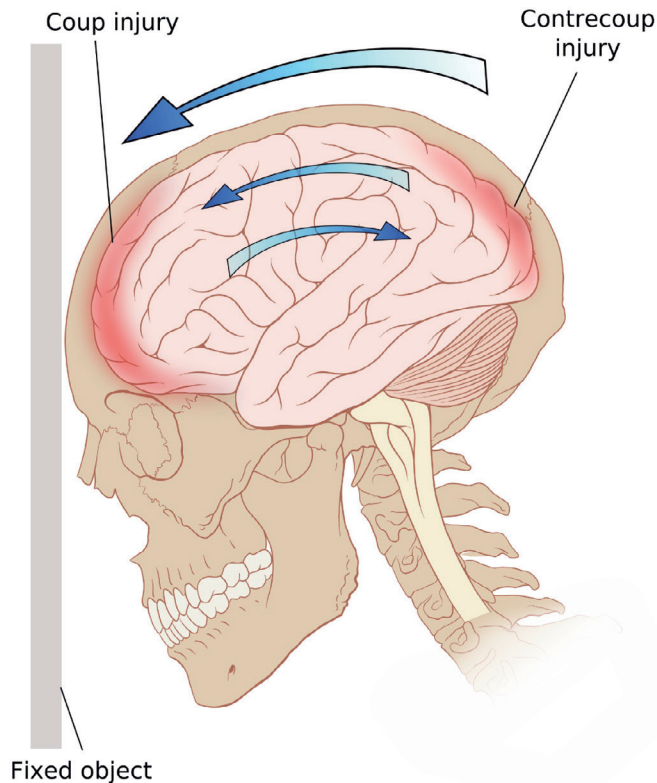


Fig. 123.1 Coup and contrecoup injuries. (By Patrick J. Lynch, medical illustrator [CC-BY-SA-3.0 (<http://creativecommons.org/licenses/by-sa/3.0/>)], via Wikimedia Commons.)

of a concussion reflect functional dysregulation instead of structural injury, concussion cannot be diagnosed with computed tomography (CT) or magnetic resonance imaging (MRI).^{8,15} However, imaging is used to rule out more serious injury, mainly hemorrhage.

History

Detailed history is imperative in patients sustaining head injury. Physicians should first determine whether further emergent neuroimaging is needed to rule out more severe brain injuries. Moreover, it is important to assess a patient's mental status and any changes since the initial injury. In many cases, it may be necessary to obtain a medical history from parents, friends, teammates, and coaches. In patients with concussions specifically, initial presentation may include a variety of symptoms and with differing severity. Physical symptoms can include headache, nausea, vomiting, issues with balance, photophobia, numbness and tingling, dizziness, and tinnitus. Cognitive symptoms may include depression, fatigue, irritability, sadness, nervousness, memory problems, and difficulty concentrating.^{8,10} Furthermore, it is important to note if patients have sustained previous head injury or concussions because a prior concussion is a significant risk factor for subsequent concussions.¹⁶ And although greater than 90% of sports-related concussions among children are not associated with an LOC, it is important to inquire about LOC to rule out other injuries.^{17,18} Symptoms after the first 4 weeks after injury are considered those of the initial concussion, and those after 4 weeks are considered postconcussion syndrome (PCS),

which can last anywhere from weeks to several months. It is therefore important to obtain pertinent history from both the initial injury and at follow-up to determine if any new or different symptoms are present and if they are part of normal recovery. A large majority (80% to 90%) of concussions resolve in 7 to 10 days; however, full recovery can vary, particularly in children and adolescents.⁸

Physical Exam

An onsite licensed health care provider should initially ensure that the athlete is medically stable and has an open airway, adequate breathing, and circulation (ABCs). The health care provider must then determine the appropriate method of removing the player from the field and address any first aid issues such as active bleeding. After this, one should observe the injured player for immediate clinical symptoms and physical signs such as LOC, headache, dizziness, drowsiness, confusion, and seizure. Physicians, if possible, should also assess gait, balance, cranial nerve function, and coordination as part of a comprehensive neurologic examination. If the athlete displays an altered mental status or LOC, he or she should be immediately transferred to a hospital with neurosurgical services.

To assess a potentially concussive injury, one can use a sideline assessment tool such as the third edition of the Sport Concussion Assessment Tool (SCAT3).¹⁹ The SCAT3 is a standardized checklist for evaluating injured athletes for concussion, incorporating the Glasgow Coma Scale, Maddocks Score, scaled score of physical symptoms, cognitive assessment, neck examination, balance examination, and coordination examination. After initial assessment, the player should be monitored for increasing severity of initial symptoms or emergence of new symptoms. Those diagnosed with concussion should not be allowed to return to play (RTP) on the day of injury.⁸

Imaging

As mentioned previously, concussions do not result in structural damage to the brain and therefore cannot be distinguished with CT or MRI. However, if there is a suspected intracranial hemorrhage from athletic injury, CT should be used. It is important for on-field medical staff and personnel to understand the general characteristics of the four basic types of hematomas: epidural, subdural, subarachnoid, and intracerebral.

Epidural hematomas are typically, but not always, characterized by a brief LOC. Individuals might present with a lucid interval where there is marked improvement before rapid decline in mental status. The athlete may regain consciousness before the subsequent deterioration and headache. This occurs due to blood from meningeal arteries building up between the dura mater and the skull. As a result a convex collection of blood is noted on imaging. Bleeding usually occurs on the same side of injury, and players who sustain epidural hematomas should be removed from the field and escorted to a hospital with neurosurgical services. There, the hematoma must be evacuated and a full recovery is expected if treated promptly.²⁰

A subdural hematoma is the most common fatal athletic head injury. Injured players experience LOC and do not usually regain consciousness. They should be sent immediately for neurosurgical

evaluation. Blood escapes from torn bridging veins in the subdural space and builds between the dura mater and arachnoid mater, ultimately resulting in a concave collection of blood expanding past suture lines.²¹ Although a smaller subdural hematoma is self-limiting and requires only small catheter drill for evacuation, a craniotomy is needed to access a larger subdural hematoma. The risk of complications is dependent on the size and rate of change of the subdural hematoma, but the prognosis is poor because of direct damage to the brain.²²

A subarachnoid hemorrhage (SAH) consist of bleeding into the subarachnoid space, an area that houses CSF on the surface of the brain. A player who suffers an SAH from head trauma will likely present with rapid onset, severe headache (“worst headache ever”), pulsating towards the back of the head. Some patients will have neck stiffness, vomiting, seizures, and a decrease or LOC. Players should be taken to a hospital for a neurosurgical evaluation with CT scan. There they will receive treatment to reduce bleeding and prevent vasospasm.²³ Surgery is likely not required if no vascular ruptures are present and the bleeding is contained superficially.²⁴

An intracerebral hemorrhage involves bleeding of the brain tissue itself. After the initial traumatic head injury, an athlete with cerebral bleeding will often present with severe headache and vomiting. This injury is commonly confused with SAH. Players should be referred to neurosurgical services. The mortality rate is between 35% and 52% and is even higher when the brainstem is affected; only 20% of survivors of survivors are expected to have a full recovery at 6 months.²⁵

Decision-Making Principles

As with all head injuries, players should be initially assessed for airway, breathing, and circulation and have their cervical spine stabilized. Athletes should be immediately removed from play if a concussion is suspected. The diagnosis of a concussion should be made using a multidisciplinary approach involving clinical symptoms, physical signs, cognitive impairment, neurobehavioral features, and sleep disturbance.⁸ It might be necessary to gather information from families, coaches, and teammates if a player is unable to provide enough details of the injury. Furthermore, clinical assessment tools such as Standardized Assessment of Concussion (SAC), Balance Error Scoring System (BESS), and SCAT3¹⁹ can be used but should not supplant a clinician’s judgment in assessing an athlete for a concussion. Because concussions are the result of functional but not structural damage to the brain, neuroimaging contributes little to the diagnosis and should only be used to rule out intracranial bleeding. Thus clinicians should avoid ordering a brain CT or brain MRI unless there is concern for a skull fracture, progressive neurologic symptoms, or focal neurologic findings.²⁶

Treatment Options

The only proven treatments for concussions are cognitive and physical rest. Cognitive rest entails avoiding concentration-driven activities such as school work, video games, and even leisurely reading. Players diagnosed with concussions should not RTP on the same day of injury; however, there is no broad consensus on the appropriate guidelines for returning players. It is widely agreed

that players should not RTP until they are asymptomatic,^{8,16,27} and during this recovery period, players should not take analgesic or sedative medications to mask postconcussion symptoms.²⁸ The latest (2016) Zurich International Consensus Statement on Concussion⁸ in Sport and the American Medical Society for Sports Medicine¹⁶ both endorse a stepwise progression in physically demanding activity for players who previously suffered from concussion and are subsequently asymptomatic off medicine. The steps of the Zurich Statement progress in the following order: symptom-limited activity, light aerobic exercise, sport-specific exercise, noncontact training drills, full-contact practice, and finally RTP. Each of these steps must be separated by at least 24 hours of asymptomatic activity before the athlete may progress to the next step after a brief initial period of rest (24–48 hours). Finally, it is generally recommended that children suffering from concussions be treated more conservatively than adults.^{8,27} Athletes with concussions should be monitored for symptoms while in the rest and recovery period. Individuals who suffer concussions are at increased risk of suffering subsequent concussions.^{29–31}



Authors’ Preferred Technique

Players with suspected concussions should be immediately removed from play and assessed by a licensed health care professional for clinical symptoms and physical signs of the injury. If the player shows signs characteristic of intracranial bleeding, he or she should be quickly sent to a hospital with neurologic services and there undergo appropriate neuroimaging and surgery, if necessary. Individuals suffering only a concussion should be given time for physical and cognitive rest until symptoms resolve before returning to light aerobic physical activity. They can then graduate in a stepwise fashion to full-contact play if they remain asymptomatic while off medications.

Results

As mentioned previously, the overwhelming majority (80% to 90%) of concussions will resolve spontaneously in 7 to 10 days. Those with symptoms persisting past 10 days should be evaluated for coexisting pathologies through conventional neuroimaging and neuropsychological testing. Although there is no effective drug treatment for concussion itself, symptoms of concussions and PCS that persist can be treated with antidepressants and analgesics, as well as cognitive, vestibular, physical, and psychological therapy.³²

Complications

Postconcussion Syndrome

PCS is a set of symptoms that lasts more than 4 weeks after the initial concussion. Common symptoms include headache, dizziness, insomnia, fatigue, irritability, poor concentration, poor memory, depression, and photophobia. There is a correlation between severity of symptoms and severity of neurologic impairment.³³ Diagnosis of PCS is difficult because of the great overlap of symptoms may be observed in conditions such as chronic pain (CP).³⁴ PCS usually resolves within 1 month, but the syndrome can last from months to years and possibly cause permanent damage or disability.³⁵ Persistent postconcussive syndrome (PPCS) is used to define a set of symptoms lasting 3 or more months after the injury. Although there is no treatment for PCS,

symptoms of PCS can be treated with antidepressants or nonsteroidal analgesics. In addition, some 40% of patients with PCS are referred to psychological consultations.³² Furthermore, earlier intervention by specialists was correlated with reduced severity of postconcussion symptoms.³⁶ However, there is no reliable way to treat PCS when it persists.

Future Considerations

Despite the wide prevalence of concussions, the research on its treatment is lacking. Further inquiry is required to understand the optimal time and type of rest for recovery. Moreover, although it is thought that some low-level exercise might be of benefit for those slow to recover from concussions, the optimal timing and intensity of activity have not been established.⁸ Studies are also needed to investigate pharmacologic interventions in managing symptoms of concussions that last greater than 10 days. Finally, further research is needed into concussions as a predictor of chronic neurobehavioral impairment.³⁷

SPINE

Introduction

The human spinal column is made up of 33 vertebrae. Each vertebra, composed of bone and hyaline cartilage, consists of a vertebral body and vertebral arch, formed by laminae and pedicles. The spine functions to provide support the body and protect the spinal cord within the vertebral foramen. Spinal nerves exit and enter through intervertebral foramina for the spine to serve as the bridge between the central and peripheral nervous systems.

Approximately 9% of spinal cord injuries (SCIs) in the United States are caused by sporting accidents, making them the fourth leading cause of SCI, behind motor vehicle accidents, falls, and violence.^{38–40} Common spine injuries include muscle spasms, avulsion fractures, compression fractures, disc herniations, and strains. More severe, although less common, are catastrophic cervical spine injuries (CCSIs), which are defined as injuries that cause structural damage to the cervical spine and are associated with actual or potential damage to the spinal cord. CCSIs include unstable fractures or dislocations, intervertebral disc herniations, and cervical cord neuropraxia (CCN). This chapter will focus on the most common mild sports spine injury, muscle strains and ligamentous sprain, and the most frequent CCSI, unstable fractures and dislocations.⁴¹

UNSTABLE FRACTURES AND DISLOCATIONS OF THE CERVICAL SPINE

A fracture and/or dislocation is defined as unstable if, when placed under a physiologic load, the spine is unable to maintain the same preinjury motions to prevent major deformities, debilitating pain, or damage to the spinal cord and nerve roots.⁴² The historical radiographic criteria for unstable fractures and dislocations are 11 degrees of angular displacement or 3.5 mm of horizontal displacement between adjacent vertebrae.⁴³ Unstable fractures and dislocations of the cervical spine typically result in permanent neurologic damage. These injuries usually occur in the lower cervical spine, where the spinal cord occupies nearly 75%

of the cross-sectional area of the spinal canal.⁴⁴ In high-contact sports such as hockey and football, the injury most frequently associated with cervical spine injury is compression. In its neutral position, the cervical spine forms a protective lordotic curve, which allows the energy impacts on cervical spine to dissipate across intervertebral discs and paravertebral muscles. However, when the head is tilted forward and flexed during impact on the top of the head, forces are transmitted down an aligned cervical spine in a process known as axial loading. If pushed passed its maximal load, the cervical spine will fail, leading to fracture, dislocation, or subluxation.

History

Obtaining a comprehensive history is important in diagnosing problems from pain in the neck and back. Firstly, identify the events surrounding the pain; high-impact injuries such as collisions or falls would be characteristic of disruptions in bones or ligaments, whereas overuse would be indicative of muscle strains or spasms. Physicians should distinguish the location of pain, the pain severity, onset and frequency of symptoms, modifying factors, and whether or not the pain radiates. Furthermore, it is necessary to conduct a thorough neurologic examination by checking for headache, impaired vision, impaired fine motor control, imbalance, impaired gait, or tingling and numbness in the extremities.

Physical Examination

It is especially important that when dealing with players with cervical spine injuries that the leading on-staff health personnel address ABCs while simultaneously immobilizing the cervical spine through manual traction or cervical orthosis. After the cervical spine has been immobilized, the injured athlete should place onto a spine board with extreme prudence. It is recommended that a player's shoulder pads and helmet should be left on until the player is moved to a hospital setting so as to minimize movement of the cervical spine.^{45,46} Exceptions to this rule includes instances where the helmet is not properly fitted to prevent independent movement of the head and helmet or if the head equipment prevents neutral alignment of the cervical spine.⁴⁶ Moreover, the helmet and shoulder pads in the cases of American football and ice hockey should not be independently removed because this will disrupt spinal alignment. Removal of the face-mask, however, is recommended and considered to be much less compromising than removal of the helmet. In cases where the airway is disrupted, jaw thrusts or head tilts can be performed. If a player's headwear in any way prevents the process of opening the airway, both the helmet and shoulder pads should be removed while maintaining cervical alignment.⁴⁷ Players with suspected cervical spine injuries should be immediately transferred to a hospital with spine trauma services.

Imaging

Indication of advanced imaging in cases of sustained blunt cervical spine trauma can be determined using the National Emergency X-Ray Utilization Study (NEXUS) clinical criteria: no midline cervical tenderness, no focal neurologic deficits, normal cognition, no intoxication, and no painful, distracting injuries. If all

five of these criteria are met, the patient has almost certainly not sustained a cervical spine injury, and further imaging is contraindicated.^{48,49} If any of the criteria are not met, x-ray films first should be taken from the lateral view so that all seven cervical vertebrae and the C7-T1 junction can be visualized. Further imaging from the lateral view with the patient's arm extended over the head (swimmer's view) might be necessary to visualize the C7-T1 junction. If there are no obvious fractures from the lateral views, physicians should proceed with anteroposterior and open-mouth odontoid views. If no fractures are observed, lateral flexion and extension views can also be taken when there is a suspected instability. If any fractures are observed in radiographs or the patient presents with neurologic deficits, CT and MRI are recommended. CT scans in the sagittal, coronal, or axial plane can provide a more detail to identify spinal fractures or dislocations (Fig. 123.2). MRIs can be used to identify damage to the spinal cord, intervertebral discs, or posterior ligamentous complex.

Decision-Making Principles

Most cervical injuries will include ligamentous sprain and/or muscle strain without osseous or neurologic injury. Players will present with pain localized on the cervical spine, reduced range of motion (ROM), and tenderness to palpitation. In these cases the injury is usually self-limited and players should be able to RTP if he or she has full ROM, full-strength in the neck, and no pain in the neck upon axial loading.⁵⁰ However, ligamentous injury can result in cervical spine instability. Severe forms of ligamentous instability in the cervical spines of young athletes are often masked by youthful ligamentous laxity, as well as through cervical muscle spasms.⁵⁰ Thus whenever subluxation of the



Fig. 123.2 Sagittal computed tomography demonstrating a cervical fracture and dislocation at the level of C6-C7. (By Frank Gaillard [CC BY-SA 3.0 (<https://creativecommons.org/licenses/by-sa/3.0/>)], via Wikimedia Commons.)

cervical spine is observed in imaging, that player should wear a cervical collar for 2 to 4 weeks before repeated imaging. Athletes should then be allowed to RTP only if there is a normal return to function without pain and no evidence of progression.

Those patients who are conscious, responsive, and not intoxicated, with suspected unstable cervical spine fractures and dislocations should be reduced as soon as possible and stabilized with a halo ring device. An MRI or contrast-enhanced CT scan can then be used to rule out the presence of retropulsed intervertebral disc material, which is implicated in the neurologic deterioration of patients undergoing reduction of the cervical spine. Athletes who sustain unstable lower cervical spine fractures or dislocations can present with a spectrum of neurologic dysfunction, with the most severe case being complete quadriplegia, whereas there is rarely spinal cord damage due to fractures or dislocations of the upper cervical spine because there is proportionally greater space within the spinal canal compared with lower segments. Furthermore, unstable cervical fractures/dislocations do not always result in upper motor neuron lesions. Under some circumstances, there will only be lower motor neuron dysfunction.

Treatment Options

Musculoligamentous sprain and strain are typically self-limited and relieved by rest, although icing and nonsteroidal antiinflammatory drugs (NSAIDs) can help to reduce pain.⁵¹ Treatment of cervical spine injury is dependent on the degree of instability and/or neurologic impairment. Unstable fractures are treated surgically with an anterior decompression and fusion or posterior spinal fusion. However, posterior stabilization and fusion was found to lead to unsuccessful outcomes in nearly half of cases of rotationally unstable cervical due to late kyphosis caused by disc collapse or inability of the midline stabilization procedure to control rotational instability.⁵² However, the anterior fusion procedure was not associated with residual deformity, and even those with incomplete spinal cord lesions and radiculopathy showed improvement in spinal cord function.⁵²



Authors' Preferred Technique

Players who sustain musculoligamentous sprain and/or strain should be removed from play and treated conservatively with rest, icing, and NSAIDs until asymptomatic. It is imperative to perform a comprehensive physical exam and history so as to rule out more serious osseous or neurologic injuries. Most sprains and strains are self-limiting, and players should be able to return to competition if they are free of neck pain upon axial loading, have a full ROM, and normal neck strength.

Athletes who sustain cervical spine fractures should at the very least be deferred from playing until the fracture has healed. If the fracture is stable, it should resolve without the need for surgical intervention and the player should be able to return by the next season. Those with unstable fractures who require a halo vest or surgical stabilization may be unable to safely return to contact sports. Moreover, those with previous cervical spine fractures are at increased risk for subsequent fractures due to

their altered spinous biomechanics. Although there is a high frequency of cervical spine injuries in contact sports, there is no universal agreement of standard guidelines for returning to pre-injury activity levels.^{53,54} Despite a lack of consensus on specific measures, it is agreed that before returning to play, athletes should be asymptomatic, pain free, neurologically intact, and have full strength and ROM.^{50,54}

Postoperative Management

Players with only musculoligamentous pain and/or strain should be monitored for deterioration of structural and neurologic function and should not engage in sporting activity until asymptomatic. Those with unstable cervical fractures and/or dislocations that require anterior surgical fusion should be monitored for degeneration of intervertebral discs adjacent to the spot of fusion.⁵⁵ Moreover, it might be necessary to stabilize the cervical spine with a cervical orthosis or halo ring device until the neck has regained its support.

Results

Ultimately, most athletes with neck or back pain have a self-limiting musculoligamentous injury that will resolve with rest and treatment with NSAIDs and icing. Athletes who sustain unstable cervical fractures and/or dislocations can lead to a wide range of results depending on severity and location. Injuries can vary from transient radiculopathies to permanent, complete SCI. The most common pattern is pronounced weakness in the upper extremities and less pronounced weakness in the lower extremities, a phenomenon known as central cord syndrome.³⁸

Complications

Disc Degeneration

Several clinical studies have pointed to increased rates of disc degeneration adjacent to the levels of anterior cervical spinal fusion.^{55–57} This is caused by the shift of load from fused spinal segment to adjacent levels, causing earlier disc degeneration. Moreover, significant increases in intradiscal pressure and segmental motion occurs at levels adjacent to disc fusion during normal ranges of motion.⁵⁵

Future Considerations

Unstable cervical spine fractures, although rare, are among the most disabling injuries in contact sports. Further research is needed into developing better protective equipment, as well as educating players, coaches, and sporting administrators into preventing maneuvers that put the cervical spines of athletes at

risk. Finally, more scientific inquiries should be made into the variety of surgical treatments for unstable cervical spine fractures, particularly comparisons between anterior and posterior approaches to stabilization and fusion.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

McCroory P, et al. Consensus statement on concussion in sport: the 4th International Conference on Concussion in Sport held in Zurich, 2012. *Br J Sports Med*. 2013;47(5):250–258.

Level of Evidence:

IV

Summary:

This paper revises and updates the consensus formed in the first three symposia on concussion in sport. It is intended for use by health care professionals in dealing with patients suffering from sports-related concussions. The document addresses issues such as the definition of concussion, concussion management, return-to-play protocols, and injury prevention. Finally, it incorporates the SCAT3 assessment tool.

Citation:

Harmon KG, et al. American Medical Society for Sports Medicine position statement: concussion in sport. *Br J Sports Med*. 2013;47(1):15–26.

Level of Evidence:

IV

Summary:

This document is presented as a tool for physicians to evaluate and manage sports-related concussions. It includes discussions of the definition, pathophysiology, incidence, diagnosis, education, prevention, and legislation of concussions in the United States.

Citation:

Cantu RC, et al. Return to play after cervical spine injury in sports. *Curr Sports Med Rep*. 2013;12(1):14–17.

Level of Evidence:

III

Summary:

The authors of this paper present detailed descriptions on a variety of cervical spine injuries in sports. It includes discussions and return to play protocols for burners or stingers, spear tackler's spine, muscular sprain and ligamentous strain, cervical spine fractures, spinal stenosis, and transient quadriplegia.

REFERENCES

1. Deleted in review.
2. Deleted in review.
3. Deleted in review.
4. Drew LB, Drew WE. The contrecoup-coup phenomenon. *Neurocrit Care*. 2004;1(3):385–390.
5. Deleted in review.
6. Deleted in review.
7. Cloots RJH, et al. Multi-scale mechanics of traumatic brain injury: predicting axonal strains from head loads. *Biomech Model Mechanobiol*. 2013;12(1):137–150.
8. McCrory P, Meeuwisse W, Dvořák J, et al. Consensus statement on concussion in sport—the 5 international conference on concussion in sport held in Berlin, October 2016. *Br J Sports Med*. 2017;51(11):838–847.
9. Ropper AH. Athletic concussion: current understanding as of 2007. *Neurosurgery*. 2007;61(6):E1340.
10. Meehan WP, Bachur RG. Sport-related concussion. *Pediatrics*. 2009;123(1):114–123.
11. Meier TB, et al. The underreporting of self-reported symptoms following sports-related concussion. *J Sci Med Sport*. 2015;18(5):507–511.
12. Bramley H, et al. High school soccer players with concussion education are more likely to notify their coach of a suspected concussion. *Clin Pediatr (Phila)*. 2011;51(4):332–336.
13. Register-Mihalik JK, et al. Knowledge, attitude, and concussion-reporting behaviors among high school athletes: a preliminary study. *J Athl Train*. 2013;48(5):645–653.
14. McCrea M, et al. Immediate neurocognitive effects of concussion. *Neurosurgery*. 2002;50(5):1032–1042.
15. McCrory P, et al. Can we manage sport related concussion in children the same as in adults? *Br J Sports Med*. 2004;38(5):516–519.
16. Harmon KG, et al. American Medical Society for Sports Medicine position statement: concussion in sport. *Br J Sports Med*. 2013;47(1):15–26.
17. Cantu RC. An overview of concussion consensus statements since 2000. *Neurosurg Focus*. 2006;21(4):1–6.
18. Halstead ME, Walter KD. Sport-related concussion in children and adolescents. *Pediatrics*. 2010;126(3):597–615.
19. SCAT3. *Br J Sports Med*. 2013;47(5):259.
20. Smith SW, et al. Emergency department skull trephination for epidural hematoma in patients who are awake but deteriorate rapidly. *J Emerg Med*. 2010;39(3):377–383.
21. Gennarelli TA, Thibault LE. Biomechanics of acute subdural hematoma. *J Trauma Acute Care Surg*. 1982;22(8):680–686.
22. Mori K, Maeda M. Surgical treatment of chronic subdural hematoma in 500 consecutive cases: clinical characteristics, surgical outcome, complications, and recurrence rate. *Neurol Med Chir (Tokyo)*. 2001;41(8):371–381.
23. Fisher CM, Kistler JP, Davis JM. Relation of cerebral vasospasm to subarachnoid hemorrhage visualized by computerized tomographic scanning. *Neurosurgery*. 1980;6(1):1–9.
24. Mayberg MR, et al. Guidelines for the management of aneurysmal subarachnoid hemorrhage. *Circulation*. 1994;90(5):2592–2605.
25. Caceres JA, Goldstein JN. Intracranial hemorrhage. *Emerg Med Clin North Am*. 2012;30(3):771–794.
26. Yuh EL, Hawryluk GW, Manley GT. Imaging concussion: a review. *Neurosurgery*. 2014;75:S50–S63.
27. Giza CC, et al. Summary of evidence-based guideline update: evaluation and management of concussion in sports: report of the Guideline Development Subcommittee of the American Academy of Neurology. *Neurology*. 2013;80(24):2250–2257.
28. West TA, Marion DW. Current recommendations for the diagnosis and treatment of concussion in sport: a comparison of three new guidelines. *J Neurotrauma*. 2014;31(2):159–168.
29. Guskiewicz KM, et al. Cumulative effects associated with recurrent concussion in collegiate football players: the NCAA Concussion Study. *JAMA*. 2003;290(19):2549–2555.
30. Guskiewicz KM, et al. Recurrent concussion and risk of depression in retired professional football players. *Med Sci Sports Exerc*. 2007;39(6):903–909.
31. Colvin AC, et al. The role of concussion history and gender in recovery from soccer-related concussion. *Am J Sports Med*. 2009;37(9):1699–1704.
32. Mittenberg W, et al. Treatment of post-concussion syndrome following mild head injury. *J Clin Exp Neuropsychol*. 2001;23(6):829–836.
33. King NS. Post-concussion syndrome: clarity amid the controversy? *Br J Psychiatry*. 2003;183(4):276–278.
34. Smith-Seemiller L, et al. Presence of post-concussion syndrome symptoms in patients with chronic pain vs mild traumatic brain injury. *Brain Inj*. 2003;17(3):199–206.
35. Ryan LM, Warden DL. Post concussion syndrome. *Int Rev Psychiatry*. 2003;15(4):310–316.
36. Wade DT, et al. Routine follow up after head injury: a second randomised controlled trial. *J Neurol Neurosurg Psychiatry*. 1998;65(2):177–183.
37. Gavett BE, Stern RA, McKee AC. Chronic traumatic encephalopathy: a potential late effect of sport-related concussive and subconcussive head trauma. *Clin Sports Med*. 2011;30(1):179–xi.
38. Maroon JC, et al. Central cord syndrome. *Clin Neurosurg*. 1991;37:612–621.
39. Sekhon LHS, Fehlings MG. Epidemiology, demographics, and pathophysiology of acute spinal cord injury. *Spine*. 2001;26(24S):S2–S12.
40. Lee BB, et al. The global map for traumatic spinal cord injury epidemiology: update 2011, global incidence rate. *Spinal Cord*. 2014;52(2):110–116.
41. Boden BP, Jarvis CG. Spinal injuries in sports. *Neurol Clin*. 2008;26(1):63–78.
42. Banerjee R, Palumbo MA, Fadale PD. Catastrophic cervical spine injuries in the collision sport athlete, part 1: epidemiology, functional anatomy, and diagnosis. *Am J Sports Med*. 2004;32(4):1077–1087.
43. White AA 3rd, et al. Biomechanical analysis of clinical stability in the cervical spine. *Clin Orthop Relat Res*. 1975;109:85–96.
44. Parke WW. Correlative anatomy of cervical spondylotic myelopathy. *Spine*. 1988;13(7):831–837.
45. Waninger KN, Swartz EE. Cervical spine injury management in the helmeted athlete. *Curr Sports Med Rep*. 2011;10(1):45–49.
46. Swartz EE, et al. National athletic trainers' association position statement: acute management of the cervical spine-injured athlete. *J Athl Train*. 2009;44(3):306–331.
47. Cantu RV, Cantu RC. Cervical spine injuries in sports. In: O'Brien M, Meehan WP III, eds. *Head and Neck Injuries in Young Athletes*. Cham: Springer International Publishing; 2016:143–156.
48. Hoffman JR, et al. Validity of a set of clinical criteria to rule out injury to the cervical spine in patients with blunt trauma.

- National Emergency X-Radiography Utilization Study Group. *N Engl J Med*. 2000;343(2):94–99.
49. Hoffman JR, et al. Selective cervical spine radiography in blunt trauma: methodology of the National Emergency X-Radiography Utilization Study (NEXUS). *Ann Emerg Med*. 1998;32(4):461–469.
 50. Cantu RC, et al. Return to play after cervical spine injury in sports. *Curr Sports Med Rep*. 2013;12(1):14–17.
 51. Young S, Aprill C, Laslett M. Correlation of clinical examination characteristics with three sources of chronic low back pain. *Spine J*. 2003;3(6):460–465.
 52. Lifeso RM, Colucci MA. Anterior fusion for rotationally unstable cervical spine fractures. *Spine*. 2000;25(16):2028–2034.
 53. Vaccaro AR, et al. Return to play criteria for the athlete with cervical spine injuries resulting in stinger and transient quadriplegia/paresis. *Spine J*. 2002;2(5):351–356.
 54. Huang P, et al. Return-to-play recommendations after cervical, thoracic, and lumbar spine injuries. *Sports Health*. 2016;8(1):19–25.
 55. Eck JC, et al. Biomechanical study on the effect of cervical spine fusion on adjacent-level intradiscal pressure and segmental motion. *Spine*. 2002;27(22):2431–2434.
 56. Katsuura A, et al. Kyphotic malalignment after anterior cervical fusion is one of the factors promoting the degenerative process in adjacent intervertebral levels. *Eur Spine J*. 2001;10(4):320–324.
 57. Goffin J, et al. Long-term follow-up after interbody fusion of the cervical spine. *J Spinal Disord Tech*. 2004;17(2):79–85.